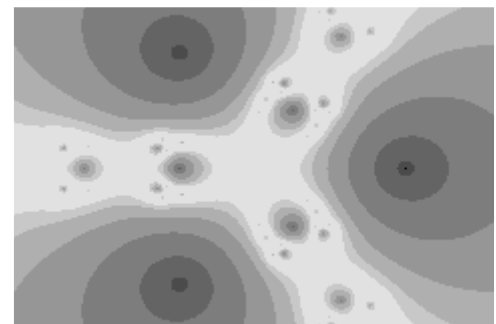


Task 4. Newton-Raphson Fractal

Available marks: 23

Fractals are exquisitely detailed images, derived from iterative algorithms, that exhibit self-similarity, that is, as you zoom in to parts of the image the same kind of structures occur, without limit. Fractals are normally generated in the complex plane, but you don't need to understand complex arithmetic to produce them, though you will have to know about polar and Cartesian (X-Y) forms of points in two dimensions, and it is strongly recommended that you implement the nine short functions described below and base your solution on them.

Fractals are generated for a rectangular part of the complex plane (or equivalently, a range of x and range of y) by associating each pixel with a point on the plane and applying an algorithm that produces a small integer, call it the *intensity* of the point. The intensity is then mapped to a colour value from a table of colours and the process repeated for the entire image.



A low-resolution greyscale fractal image focusing on the complex cube roots of unity. x ranges from -1.6 to 1.6 and y from -1.2 to 1.2

For this task intensity is calculated using an elegant iterative algorithm that finds roots of equations, discovered in 1690 by Joseph Raphson (and independently by Isaac Newton). Say you have a function in one variable $f(x)$, whose derivative $f'(x)$ is well defined, and you want to find a value of x for which $f(x) = 0$. Starting with some guess x_0 , calculate

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

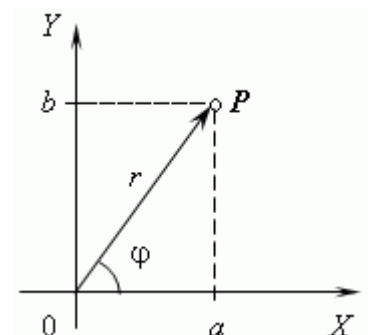
x_1 is normally closer to the root, so can be used as the next guess. Continue like this until successive guesses differ by a small amount, say 10^{-10} . For example, if $f(x) = x^3 - 1$, its real root is 1. $f'(x) = 3x^2$, so the iterative formula to generate a new guess x' from the previous guess x is

$$x' = x - (x^3 - 1) / (3x^2)$$

If you start with, say, $x = 5$, the successive approximations are 5, 3.34667, 2.26087, 1.57246, 1.18312, 1.02688, 1.0006975, 1.000000486, 1.000000000000024, 1.

Complex Arithmetic

We can apply the Newton-Raphson technique to functions of more than one variable. All that's necessary is to define and code arithmetic operations, including the polar coordinate transformations that make the arithmetic work.



Symbolically z is a point (x, y) , which in polar form has magnitude r and argument (angle with respect to the x axis) ϕ . The following 9 equations define all the required transformations for this task in terms of real arithmetic:

$\text{mag}(z) = r = \sqrt{x^2 + y^2}$ # use **sqrt** or a similarly named function
 $\text{arg}(z) = \phi = \tan^{-1}(y / x)$ # hint: use the **atan2** function if it's available, as it's defined for all x, y
 $\text{cart}(r, \phi) = (r \cos \phi, r \sin \phi)$ # Cartesian form given polar coordinates
 $z_1 + z_2 = (x_1 + x_2, y_1 + y_2)$
 $z_1 - z_2 = (x_1 - x_2, y_1 - y_2)$
 $z_1 * z_2 = \text{cart}(\text{mag}(z_1) * \text{mag}(z_2), \text{arg}(z_1) + \text{arg}(z_2))$ # Multiply magnitudes, add arguments
 $z_1 / z_2 = \text{cart}(\text{mag}(z_1) / \text{mag}(z_2), \text{arg}(z_1) - \text{arg}(z_2))$ # Divide magnitudes, subtract arguments
 $z * k = (x*k, y*k)$ # Scaling by a real value k
 $z^k = \text{cart}(\text{mag}(z)^k, \text{arg}(z)*k)$ # Exponentiation by a real value k : exponential magnitude, scaled arg

Thus for the cube root equation,

$$z' = z - (z^3 - 1) / (3 z^2)$$

Your Task

Write a program that draws the fractal representing the calculation of the complex cube roots of 1 using the Newton-Raphson method, over a specified part of the complex plane. The intensity of each point is the number of iterations before the algorithm converges. Convergence occurs if the magnitudes of successive values differ by less than 10^{-10} . The algorithm should be stopped if it requires more than N iterations, N being a parameter to the program.

The way the pixels are coloured is either

- By using a greyscale that varies uniformly from black (1 iteration) to white (N or more iterations), or
- By reading a file of RGB colours, one per line, in HTML uppercase hexadecimal form (such as #FFFFFF for white), starting with intensity 1. The number of lines in the file sets the value of N .

Assessment

The judges will ask you to generate and display a fractal image using each of the following sets of parameters. The PPM image aspect ratio must be exactly 4:3 (width:height), with the width at least 320 pixels but is otherwise up to you.

Range of x	Range of y	Colourisation	Marks available
-1.6 to 1.6	-1.2 to 1.2	18-level greyscale	8
-1.6 to 1.6	-1.2 to 1.2	Palette in file pal16.txt	4
0.41 to 0.59	0.38 to 0.50	Palette in file pal16.txt	5
0.50 to 0.56	0.46 to 0.50	Palette in file pal21.txt	6